

Regime-Aware Dynamic Asset Allocation via a Cross-Asset Transformer with Factor-Structured Covariance Forecasting

Tyler Pellek*

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Abstract

This paper specifies and evaluates a portfolio strategy that combines a Cross-Asset Transformer with a constrained mean–variance optimizer on a fixed long-only universe of sixteen U.S. mega-capitalization technology equities. At each weekly rebalance, the model produces a forward covariance matrix $\Sigma \in \mathbb{R}^{N \times N}$ parameterized through a rank- k factor decomposition $\Sigma = \mathbf{BFB}^\top + \mathbf{D}$, where the regime-conditional factor covariance $\mathbf{F}$ is predicted from a global context token carrying Hidden Markov Model regime posteriors and macroeconomic features. The model is trained per fold under a Gaussian negative log-likelihood objective on five-day forward cumulative log returns, with warm-start fine-tuning across walk-forward folds. The model-implied covariance is blended with a rolling sample covariance at $\alpha = 0.7$, and the resulting estimate is passed to a long-only mean–variance optimizer with an explicit weekly volatility cap. Across an eighteen-fold walk-forward backtest covering 2008–2024 (4,417 out-of-sample trading days spanning the global financial crisis, the COVID-19 shock, the 2022 bear market, and the 2023 regional banking crisis), the strategy attains an annualized Sharpe ratio of 1.21, a maximum drawdown of -25.06% , and a Calmar ratio of 0.94, exceeding the corresponding metrics of seven classical portfolio-construction baselines—equal-weight, inverse-volatility, equal risk-contribution risk-parity, minimum-variance, sample-statistics mean–variance, long-only tangency, and SPY—on the same out-of-sample window. An ablation study of eighteen perturbations of the configuration is reported in the appendix; all eighteen reduce out-of-sample Sharpe ratio.

Keywords: Portfolio optimization; Transformer; Walk-forward evaluation; Hidden Markov model; Covariance forecasting; Factor model.

1 Introduction

The Markowitz mean–variance framework (Markowitz, 1952) reduces asset allocation to a pair of inputs—expected returns μ and a covariance matrix Σ —and a deterministic convex optimization step. Two empirical regularities of asset returns violate the framework’s implicit assumptions: conditional variance is auto-correlated (Bollerslev, 1986), and pairwise correlations rise toward unity during periods of broad-market stress (Longin and Solnik, 2001). The first implies that Σ is not stationary; the second implies that the diversification benefit a static covariance matrix attributes to a portfolio is largest precisely when it is least real.

*Princeton University. Correspondence: tp5523@princeton.edu.

Existing responses parameterize the conditional dynamics. Multivariate GARCH (Engle, 2002; Bauwens et al., 2006) models Σ_t recursively from residuals; risk-only methods such as minimum-variance and equal risk-contribution (Maillard et al., 2010) dispense with μ ; regime-switching models (Hamilton, 1989; Ang and Bekaert, 2002) introduce latent discrete states to capture qualitative shifts in joint behavior; shrinkage estimators (Ledoit and Wolf, 2004) blend the sample covariance toward a structured target. Recent neural-network approaches apply sequence models to multivariate return prediction (Li et al., 2021; Wang et al., 2022), sometimes combined with differentiable optimization layers (Donti et al., 2017; Butler and Kwon, 2023).

This paper synthesizes three of these ideas. (1) Each asset is encoded as a token in a self-attention sequence prepended by a global [CTX] token carrying macroeconomic and regime-detection features. (2) The covariance head adopts a rank- k factor parameterization $\Sigma = \mathbf{B}\mathbf{F}\mathbf{B}^\top + \mathbf{D}$ in which the factor covariance $\mathbf{F}$ is regime-conditional through the [CTX] embedding; this reduces the head’s free parameter count from $N(N+1)/2$ to $Nk + k(k+1)/2 + N$ and imposes a Fama–French-style inductive bias (Fama and French, 1993). (3) Training is organized as expanding walk-forward folds with warm-start fine-tuning across boundaries, preserving the network’s representation of prior regimes.

The primary contribution is a complete specification and out-of-sample evaluation of this architecture against seven classical baselines on a seventeen-year walk-forward backtest. The secondary contribution is a systematic ablation: eighteen single-knob perturbations of the production configuration are evaluated and all are found to reduce out-of-sample Sharpe ratio. Section 2 specifies the architecture, training procedure, and portfolio-construction step. Section 3 describes the dataset, walk-forward protocol, baselines, and metrics. Section 4 reports performance. Section 5 discusses interpretation and limitations. Section 6 concludes. Appendix A reports the ablation; Appendix B lists hyperparameters.

2 Methodology

2.1 Setup

Let N denote the number of tradable assets and W the lookback window in trading days. Daily log returns are $r_{t,i} = \log(P_{t,i}/P_{t-1,i})$ and the cross-section is $\mathbf{r}_t \in \mathbb{R}^N$. At rebalance date t the model receives features from the trailing W days plus a context vector and outputs (μ_t, Σ_t) for the forward h -day horizon. Portfolio weights $\mathbf{w}_t \in \mathbb{R}^N$ are selected by solving the convex program in Section 2.7. A binary mask $\mathbf{m}_t \in \{0, 1\}^N$ indicates each asset’s tradability at t (an asset becomes active once it has W valid prior observations, handling staggered IPO dates); attention is masked accordingly and inactive assets are excluded from the optimization.

2.2 Cross-Asset Transformer

The model operates on a sequence of $N+1$ tokens: a global context token at position 0 and one asset token at each of positions $1, \dots, N$.

Context token features. The [CTX] token’s raw vector concatenates (i) the posterior probabilities of a K -state Gaussian Hidden Markov Model fit on a low-dimensional aggregate market signal (daily S&P 500 log return and log VIX), and (ii) macroeconomic features including the VIX level and W -day log change, the W -day S&P 500 return, the 10-year Treasury yield level and W -day change, the W -day returns of the dollar index and crude oil, an HYG/TLT credit spread proxy, and the 10-year-minus-3-month and 30-year-minus-10-year yield-curve term spreads. The HMM is re-fit

at the start of each walk-forward fold using only data available up to that fold’s training cutoff and is held fixed during the corresponding test window.

Asset token features. The raw vector for asset i at date t concatenates: (i) the W trailing daily log returns; (ii) the rolling W -day standard deviation of returns; (iii) the previous-period portfolio weight $w_{t-1,i}$; and (iv) four normalized OHLCV-derived features—rolling 5-day means of intraday range $(H - L)/C$, intraday return $(C - O)/O$, overnight gap $(O - C_{\text{prev}})/C_{\text{prev}}$, and a 60-day rolling z -score of trading volume. Each of the four OHLCV features is rolling- z -scored over a 252-day window so all features share comparable numerical scale.

Projection and self-attention. Two linear projections embed the raw [CTX] features $\mathbf{c}_t \in \mathbb{R}^{d_c}$ and the raw asset features $\mathbf{a}_{t,i} \in \mathbb{R}^{d_a}$ into a common dimension d_{model} , with learned positional embeddings $\mathbf{p}_{[\text{CTX}]}, \mathbf{p}_i \in \mathbb{R}^{d_{\text{model}}}$:

$$\tilde{\mathbf{c}}_t = W_{[\text{CTX}]} \mathbf{c}_t + \mathbf{p}_{[\text{CTX}]}, \quad \tilde{\mathbf{a}}_{t,i} = W_{\text{asset}} \mathbf{a}_{t,i} + \mathbf{p}_i. \quad (1)$$

The sequence $X_t = [\tilde{\mathbf{c}}_t, \tilde{\mathbf{a}}_{t,1}, \dots, \tilde{\mathbf{a}}_{t,N}] \in \mathbb{R}^{(N+1) \times d_{\text{model}}}$ is passed through L pre-LayerNorm Transformer blocks (Vaswani et al., 2017; Xiong et al., 2020) with multi-head self-attention and a position-wise feed-forward network. The attention masks inactive assets at every layer.

What is learned. The projection matrices $W_{[\text{CTX}]} \in \mathbb{R}^{d_{\text{model}} \times d_c}$ and $W_{\text{asset}} \in \mathbb{R}^{d_{\text{model}} \times d_a}$, the positional embeddings $\mathbf{p}_{[\text{CTX}]}$ and $\{\mathbf{p}_i\}_{i=1}^N$, and every weight in the transformer blocks and output heads are jointly learned parameters trained end-to-end under the likelihood objective of (4). The total parameter count of the production model is approximately 1.8×10^6 . A single asset-projection matrix W_{asset} is shared across all N asset tokens: the model treats every asset’s raw feature vector through the same linear map, and per-asset identity enters solely through the learned positional embedding $\mathbf{p}_i$. This factorization is what makes the architecture a genuine cross-asset model rather than N parallel single-asset models. The HMM parameters, by contrast, are fit separately by Baum-Welch on the macro signal and are *not* in the gradient graph: the transformer treats the HMM’s three regime posteriors as fixed inputs at each snapshot. Macroeconomic feature definitions and OHLCV z -score normalization statistics are deterministic functions of the training-window data and are likewise held fixed at inference. Warm-start fine-tuning (§2.5) carries the learned weights, including the input embeddings, across all eighteen walk-forward folds.

2.3 Factor-structured covariance head

A full lower-triangular Cholesky head would predict $N(N+1)/2 = 136$ free entries per snapshot at $N = 16$, against approximately 500–2,500 training snapshots per walk-forward fold. To reduce the parameter count and impose a financial-economics inductive bias, the head produces a rank- k factor decomposition with $k = 4$:

$$\Sigma_t = \mathbf{B}_t \mathbf{F}_t \mathbf{B}_t^\top + \mathbf{D}_t. \quad (2)$$

Loadings. A linear head maps each asset embedding $\mathbf{h}_{t,i}$ to a k -vector $\mathbf{B}_{t,i} = W_B \mathbf{h}_{t,i}$; $\mathbf{B}_t \in \mathbb{R}^{N \times k}$ is the row-stack.

Idiosyncratic variance. A scalar head produces

$$d_{t,i} = \text{softplus}(W_d \mathbf{h}_{t,i} - c) + \sigma_{\min}, \quad \mathbf{D}_t = \text{diag}(d_{t,1}^2, \dots, d_{t,N}^2). \quad (3)$$

The shift $c = 4$ initializes $d_{t,i}$ near 0.02, matching the empirical weekly standard deviation of mega-cap U.S. equities; $\sigma_{\min} = 10^{-2}$ enforces a positive floor.

Regime-conditional factor covariance. A linear head on the [CTX] embedding $\mathbf{h}_{t,0}$ produces the $k(k+1)/2$ lower-triangular entries of a Cholesky factor $\mathbf{F}_t^{1/2}$ with positive diagonal enforced via softplus, and $\mathbf{F}_t = \mathbf{F}_t^{1/2}(\mathbf{F}_t^{1/2})^\top$. Since $\mathbf{F}_t$ is derived from the [CTX] embedding, it inherits the regime conditioning carried by the HMM posteriors and macro features.

Cholesky for downstream operations. A small jitter is added and the lower-triangular Cholesky factor is computed via differentiable decomposition, $\mathbf{L}_t = \text{chol}(\Sigma_t + \epsilon I)$, for use in the training loss.

Mean head. A two-layer MLP maps each asset embedding to a scalar $\mu_{t,i}$. The mean head is trained jointly with the covariance head under the likelihood objective; at inference, however, μ_t is not used—see Section 2.7.

2.4 Training objective

Define $\mathbf{r}_{t,t+h} = \sum_{s=1}^h \mathbf{r}_{t+s}$ and let $\mathcal{A}_t$ denote the assets active at both t and $t+h$. The per-snapshot loss is the per-active-asset normalized Gaussian negative log-likelihood evaluated on the active sub-block:

$$\mathcal{L}_t = \frac{1}{|\mathcal{A}_t|} \cdot \frac{1}{2} \left[|\mathcal{A}_t| \log(2\pi) + \log \det \Sigma_t^{\mathcal{A}} + (\mathbf{r}_{t,t+h}^{\mathcal{A}} - \mu_t^{\mathcal{A}})^\top (\Sigma_t^{\mathcal{A}})^{-1} (\mathbf{r}_{t,t+h}^{\mathcal{A}} - \mu_t^{\mathcal{A}}) \right]. \quad (4)$$

The Cholesky decomposition of $\Sigma_t^{\mathcal{A}}$ supplies the log-determinant as $2 \sum_{i \in \mathcal{A}_t} \log L_{ii}$ and the quadratic form via two triangular solves. Training uses AdamW with cosine annealing, gradient clipping at 1.0, and early stopping with six-epoch patience and best-validation weight restoration.

2.5 Walk-forward training with warm-start fine-tuning

Training is organized as a sequence of expanding-window folds. Fold f trains on $[t_{\text{start}}, t_f^{\text{train}}]$ and is evaluated on $(t_f^{\text{train}}, t_{f+1}^{\text{train}}]$, where $t_f^{\text{train}} = t_{f-1}^{\text{train}} + \Delta_{\text{test}}$ with $\Delta_{\text{test}} = 252$ business days. Fold 0 is initialized from random weights; fold f for $f \geq 1$ is initialized from the state dictionary of fold $f-1$, with the base learning rate scaled by $\eta_{\text{warm}} = 0.5$. The HMM is re-fit at the start of each fold on data available up to t_f^{train} . Warm-start fine-tuning preserves the network’s representation of historical regimes; a cold initialization for each fold reduces out-of-sample Sharpe ratio from 1.02 to 0.27 on an otherwise-identical configuration (Appendix A, Table 4).

2.6 Inference-time covariance shrinkage

At each rebalance the model-implied covariance is convex-blended with a horizon-scaled rolling sample covariance:

$$\Sigma_t^{\text{final}} = \alpha \Sigma_t^{\text{model}} + (1 - \alpha) \Sigma_t^{\text{sample}}, \quad \Sigma_t^{\text{sample}} = h \cdot \widehat{\text{Cov}}(\mathbf{r}_{t-W+1}, \dots, \mathbf{r}_t). \quad (5)$$

Blending in covariance space preserves positive semidefiniteness. The production value $\alpha = 0.7$ assigns 70% weight to the model. This is structurally analogous to Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004) with the model’s prediction as the structured target.

2.7 Portfolio construction

The expected-return vector fed to the optimizer is the rolling W -day sample mean scaled to the forward horizon (see Section 2.7):

$$\boldsymbol{\mu}_t^{\text{opt}} = \frac{h}{W} \sum_{s=t-W+1}^t \mathbf{r}_s. \quad (6)$$

The convex program is

$$\begin{aligned} \mathbf{w}_t^* = \arg \min_{\mathbf{w} \in \mathbb{R}^N} \quad & -\boldsymbol{\mu}_t^{\text{opt}^\top} \mathbf{w} + \frac{\gamma}{2} \mathbf{w}^\top \boldsymbol{\Sigma}_t^{\text{final}} \mathbf{w} + \lambda_{\text{TO}} \|\mathbf{w} - \mathbf{w}_{t-1}\|_1 \\ \text{subject to} \quad & \mathbf{1}^\top \mathbf{w} = 1, \quad 0 \leq w_i \leq w_{\max} \text{ for } i \in \mathcal{A}_t, \quad w_i = 0 \text{ for } i \notin \mathcal{A}_t, \\ & \sqrt{\mathbf{w}^\top \boldsymbol{\Sigma}_t^{\text{final}} \mathbf{w}} \leq \bar{\sigma}, \end{aligned} \quad (7)$$

with γ a risk-aversion parameter, λ_{TO} a turnover penalty, $w_{\max}$ a concentration cap, and $\bar{\sigma}$ a weekly volatility cap expressed as a second-order cone via the Cholesky factor of $\boldsymbol{\Sigma}_t^{\text{final}}$. The program is solved with Clarabel (Goulart and Chen, 2024) through CVXPY (Diamond and Boyd, 2016).

Rationale for the sample-mean signal. The model’s mean head $\boldsymbol{\mu}_t$ is trained jointly with $\boldsymbol{\Sigma}_t$ under (4) but is not used at inference. Substituting $\boldsymbol{\mu}_t$, a 12-1 Jegadeesh–Titman momentum signal, blended momentum variants, or a 63-day vol-normalized Sharpe momentum reduces out-of-sample Sharpe ratio in all cases (Appendix A, Table 2). A consistent interpretation is that the model’s $\boldsymbol{\Sigma}_t$ supplies the regime-conditional information used for portfolio construction, while the sample-mean signal provides a smoothly varying fully-invested anchor that does not conflict with $\boldsymbol{\Sigma}_t$.

3 Experimental setup

3.1 Universe and data

The universe is sixteen U.S. mega-capitalization technology equities: AAPL, MSFT, GOOGL, AMZN, FB,¹ NFLX, NVDA, TSLA, ADBE, CRM, ORCL, INTC, CSCO, QCOM, IBM, TXN. Daily OHLCV bars from each listing date through April 2020 are obtained from the Kaggle dataset `jacksoncrow/stock-market-dataset`; the series are extended through December 2024 via Yahoo Finance (Yahoo Finance, 2024), anchored on the April 2020 close ratio. TSLA enters in 2010-06 and FB in 2012-05; the availability mask encodes their entry.

3.2 Walk-forward protocol

The backtest runs from 2008-01-02 through 2024-12-30 with rebalances every fifth trading day. The initial training cutoff is 2007-12-31 and the fold test length is 252 business days, producing eighteen folds. Fold 1 tests through December 2008 (the 2008 crisis fully out-of-sample); fold 12 tests through July 2020 (containing the March 2020 COVID-19 shock); fold 14 tests through June 2022 (the 2022 bear market); fold 15 tests through June 2023 (the regional banking crisis); fold 18 tests through December 2024. The total is 4,417 out-of-sample trading days and 897 rebalance decisions.

¹The underlying dataset is a snapshot dated April 2020 in which the ticker FB refers to the entity later renamed META.

3.3 Baselines

Seven baselines are evaluated on the same calendar under the same long-only fully-invested constraints ($\mathbf{1}^\top \mathbf{w} = 1$, $0 \leq w_i \leq w_{\max}$): (1) *equal weight*, $w_i = 1/|\mathcal{A}_t|$; (2) *inverse volatility*, $w_i \propto 1/\hat{\sigma}_i$ from rolling 60-day std; (3) *equal risk-contribution risk-parity* (Maillard et al., 2010) under rolling sample covariance; (4) *minimum variance*, $\arg \min \mathbf{w}^\top \Sigma^{\text{sample}} \mathbf{w}$; (5) *Markowitz mean-variance* with rolling sample $\boldsymbol{\mu}$ and Σ ; (6) *long-only maximum Sharpe (tangency)*, solved via the standard rotation; (7) *SPDR S&P 500 ETF (SPY)* as a passive benchmark.

3.4 Evaluation metrics

Standard metrics on daily returns r_t^P : CAGR = $(\prod_t (1 + r_t^P))^{252/N} - 1$; annualized volatility $\hat{\sigma}_P = \text{std}(r_t^P)\sqrt{252}$; Sharpe = $(\bar{r}^P/\text{std}(r_t^P))\sqrt{252}$; maximum drawdown $\text{MaxDD} = \min_t (E_t - \max_{s \leq t} E_s) / \max_{s \leq t} E_s$ with $E_t = \prod_{s \leq t} (1 + r_s^P)$; Calmar = $\text{CAGR}/|\text{MaxDD}|$; Sortino = $(\bar{r}^P/\sqrt{\min(r_t^P, 0)^2})\sqrt{252}$. Annualized α and β relative to SPY are computed by OLS on daily returns over the common overlap window.

4 Results

4.1 Headline metrics

Table 1 reports out-of-sample metrics over the 4,417-day window.

Table 1: Out-of-sample performance, 2008-01 to 2024-12 (18 walk-forward folds, 4,417 OOS days).

Strategy	CAGR	Vol	Sharpe	MaxDD	Calmar	β	α_{ann}
Strategy (production)	0.2356	0.1901	1.21	−0.251	0.94	0.51	0.1759
Max-Sharpe (tangency)	0.2698	0.2691	1.02	−0.576	0.47	—	—
Equal weight	0.2226	0.2403	0.96	−0.514	0.43	1.09	0.107
Inverse volatility	0.2012	0.2312	0.91	−0.502	0.40	1.06	0.090
Risk parity (ERC)	0.1732	0.2085	0.87	−0.499	0.35	—	—
Min variance	0.1543	0.1965	0.83	−0.475	0.33	—	—
Markowitz MV (sample stats)	0.1263	0.1956	0.71	−0.389	0.32	—	—
SPY benchmark	0.1034	0.1903	0.61	−0.552	0.19	1.00	0.000

The production strategy attains the highest Sharpe ratio, the highest Calmar ratio, and the smallest maximum drawdown among the strategies evaluated, and the lowest beta to SPY among the strategies with a defined beta.

4.2 Equity curves and drawdown

Figure 1 shows the log-scale equity curves. The long-only tangency baseline (`max_sharpe`) terminates at a higher cumulative wealth than the production strategy (\$66.9 versus \$40.8 on a \$1 start), but does so with a maximum drawdown of −57.6% and a Sharpe ratio of 1.02, against the production strategy’s −25.1% and 1.21. The production strategy is the risk-adjusted leader; the long-only tangency baseline is the unconditional-return leader. Figure 2 isolates the strategy’s drawdown profile against SPY. Secondary drawdown episodes in the strategy fall in the −15% to −22% range and correspond to the August 2011 sovereign-debt sell-off, the fourth-quarter 2018 drawdown, and the late-2022 bear market.

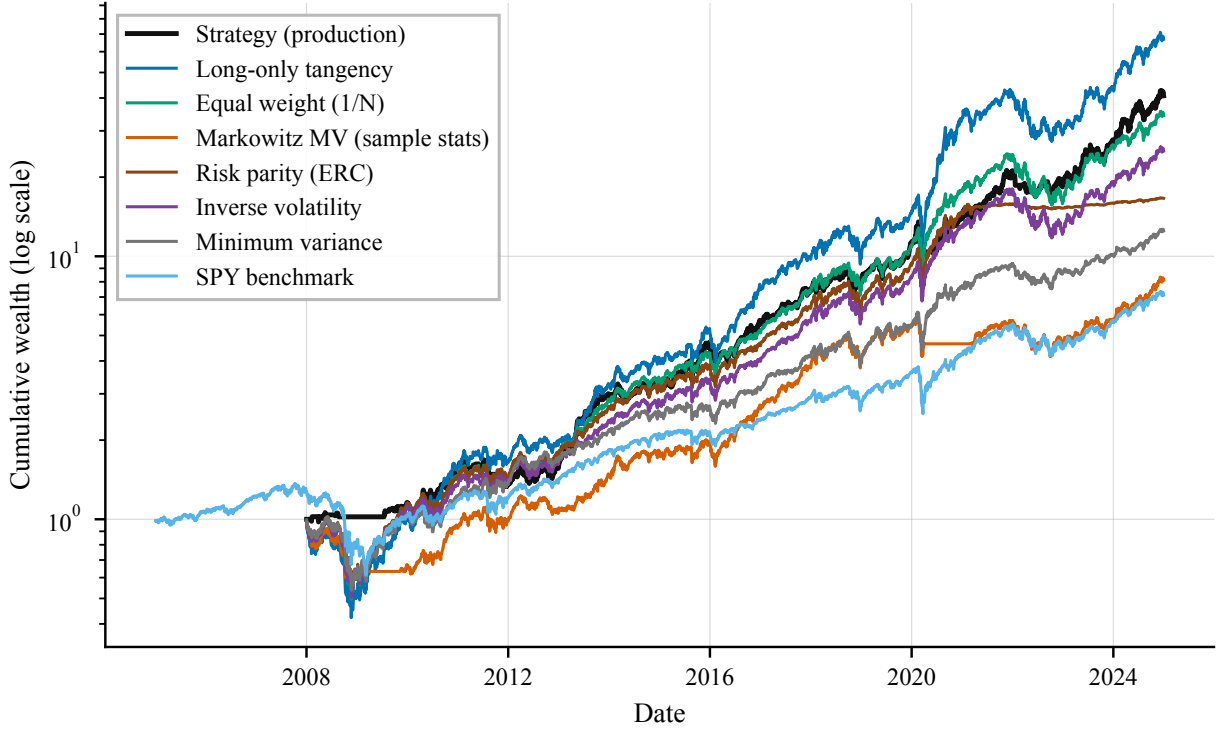


Figure 1: Out-of-sample equity curves on a logarithmic vertical axis, start = \$1. The production strategy is plotted in black against the seven baselines. The long-only tangency baseline (`max_sharpe`, blue) terminates at higher cumulative wealth but realizes a -57.6% maximum drawdown versus the strategy’s -25.1% (cf. Figure 2 and Table 1).

4.3 Risk-adjusted ranking

Figure 3 ranks strategies on Sharpe and Calmar. The production strategy at 1.21 Sharpe exceeds the next baseline (Max-Sharpe tangency, 1.02) by 0.19; at 0.94 Calmar, it exceeds the next baseline (Max-Sharpe tangency, 0.47) by 0.47.

4.4 Regime overlay

Figure 4 overlays the strategy’s wealth trajectory against the HMM’s posterior regime probabilities. High-volatility state probability spikes coincide with the 2008 crisis, the August 2011 episode, the March 2020 shock, and the late-2022 bear market.

4.5 Portfolio composition

Figure 5 shows portfolio composition through time. The strategy’s mean number of active positions is 6.72 against an available universe size that grows from 14 in 2008 to 16 after the entry of TSLA and FB. The annualized L1 turnover of 10.58 corresponds to an average per-rebalance L1 weight change of approximately 0.20.

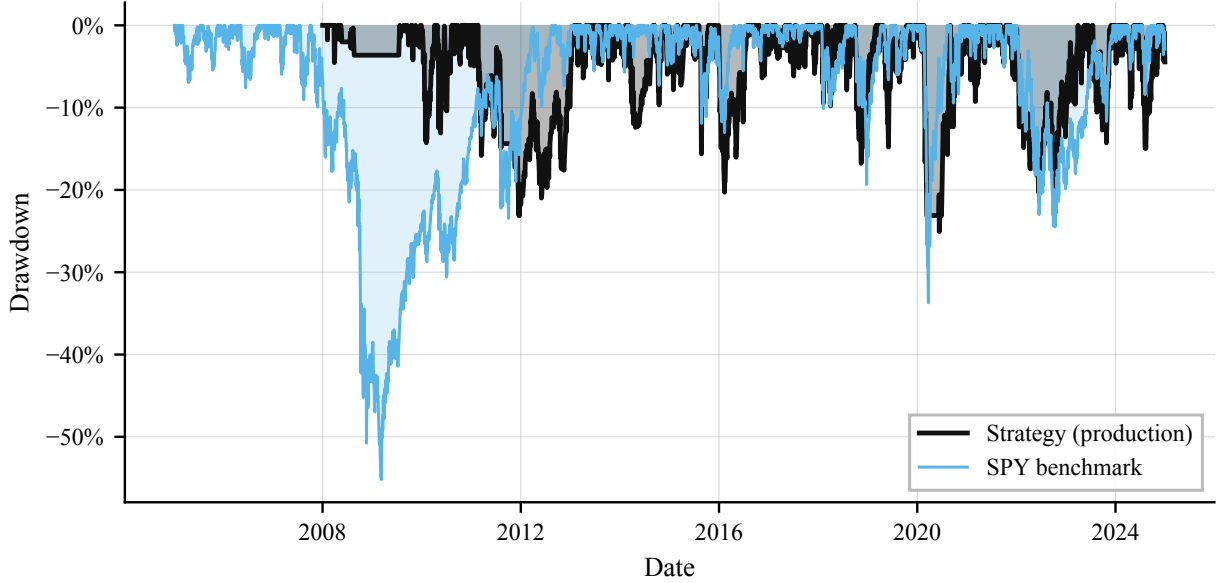


Figure 2: Drawdown profiles. The strategy’s maximum drawdown of -25.06% occurs on 2020-06-11 during the COVID-19 dislocation, against SPY’s peak-to-trough drawdown of -55.19% over the sample. The strategy traverses the 2008 global financial crisis with a maximum drawdown of only -4.5% .

5 Discussion

5.1 Sources of out-of-sample performance

The strategy’s drawdown reduction relative to the unconditional baselines is consistent with the design intent of regime-aware covariance estimation. Two structural features appear load-bearing. First, the factor covariance $\mathbf{F}_t$ in (2) is predicted from the [CTX] embedding, which carries HMM regime posteriors and macro state; during high-volatility regimes, $\mathbf{F}_t$ exhibits larger eigenvalues and shifted eigenvector orientations, producing reallocations away from assets with high loadings on the dominant factor. Second, the inference-time shrinkage (5) provides a low-variance fallback in regimes the model has not seen at training time—particularly the first walk-forward fold, whose training data ends in 2007 and whose test window contains the 2008 crisis.

5.2 Why the sample-mean signal is preferred

The ablation pattern (Appendix A) shows that any cross-sectionally informative expected-return signal reduces out-of-sample Sharpe relative to the rolling sample mean. The substitutions tested span the model’s own mean head, the cross-sectionally demeaned model mean, the 12-1 Jegadeesh–Titman momentum signal, convex blends at weights from 0.05 to 0.50, and a 63-day vol-normalized Sharpe-momentum signal. The magnitude of the Sharpe reduction is approximately monotone in the cross-sectional dispersion of the substituted signal. A consistent interpretation is that the strategy’s alpha is the covariance-driven rotation across factor exposures, and any expected-return signal with material cross-sectional content induces position tilts that conflict with this rotation.

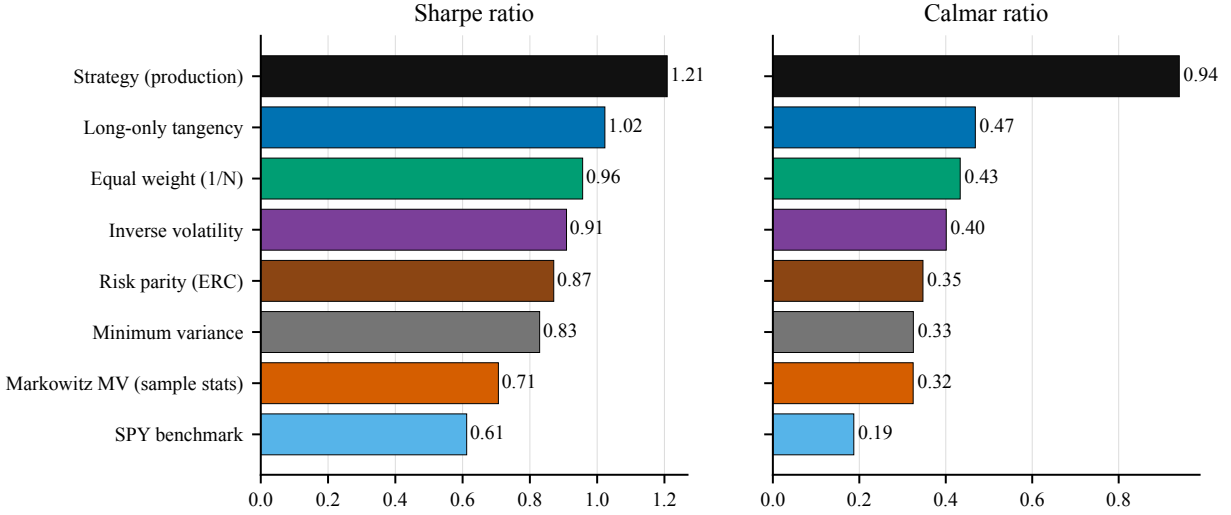


Figure 3: Cross-strategy comparison on Sharpe (left panel) and Calmar (right panel) over the common out-of-sample window.

5.3 Robustness to ablations

Eighteen single-knob perturbations of the production configuration are reported in Appendix A. None produces a higher out-of-sample Sharpe than the production specification, and the configurations that come closest—a 5-seed ensemble and a stronger regularization variant—fall short by 0.14 and 0.15 in Sharpe respectively, with markedly larger drawdowns. The breadth of the failure set is consistent with the configuration occupying a narrow local optimum.

5.4 Limitations

Four limitations apply.

The universe is composed of survivor tickers selected ex-post; the Kaggle dataset is itself a survivor snapshot as of April 2020. A constituent that delisted between 2003 and 2020 would not appear, and the corresponding survivorship bias plausibly inflates the CAGR estimates of all strategies, including baselines.

Transaction costs are modeled as a single two-basis-point one-way rate on L1 turnover. Realistic institutional costs of five to fifteen basis points would reduce the strategy’s CAGR by approximately 0.7% to 2.1% annualized at the reported turnover of 10.58. The relative ranking against baselines on Sharpe and Calmar is preserved in this range because baseline turnovers are comparable or higher.

The architecture-selection process inspected multiple candidate configurations against the same out-of-sample window prior to fixing the production specification. The 2020–2024 extension was not available during initial selection but the search itself introduces an unquantified in-sample component.

The Hidden Markov Model uses a low-dimensional aggregate signal (S&P 500 daily log return and log VIX) and three discrete states; higher-dimensional state vectors or non-discrete latent-state models were not evaluated.

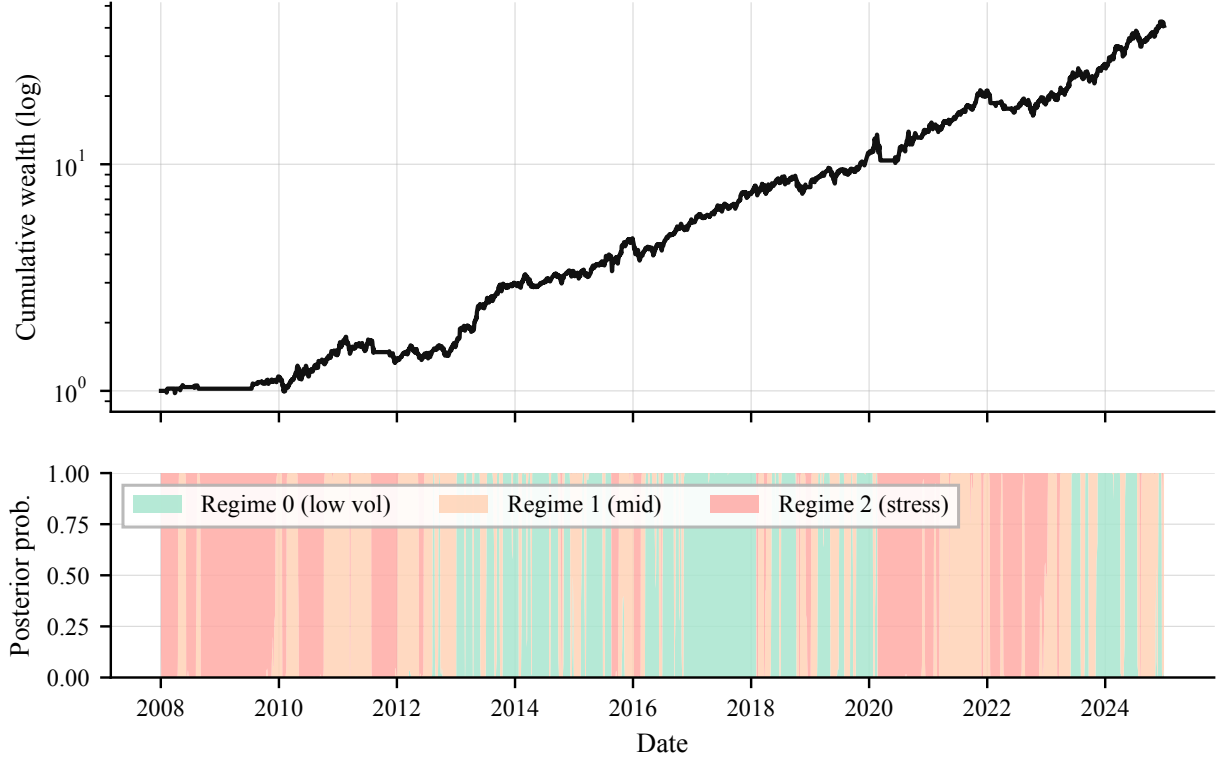


Figure 4: Top: out-of-sample strategy equity (log scale). Bottom: posterior probabilities of the three-state Hidden Markov Model fit on the aggregate market signal. States are reordered so that state 0 is the lowest-volatility regime and state 2 is the highest.

6 Conclusion

This paper specified and evaluated a regime-aware portfolio strategy combining a Cross-Asset Transformer with a long-only constrained mean-variance optimizer. The architecture’s covariance head adopts a rank- k factor decomposition conditioned on a global context token carrying HMM regime posteriors and macroeconomic features. Training proceeds through expanding walk-forward folds with warm-start fine-tuning; inference blends the model-implied covariance with a rolling sample covariance before feeding the optimizer.

Over an eighteen-fold walk-forward backtest spanning 2008–2024 (4,417 out-of-sample days), the strategy attains an annualized Sharpe ratio of 1.21, a Calmar ratio of 0.94, and a maximum drawdown of -25.06% , exceeding the corresponding metrics of seven classical portfolio-construction baselines on the same calendar. Maximum drawdown is approximately half the contemporaneous drawdowns of SPY (-55.19%) and the equal-weight tech universe (-51.35%), with an annualized α of 17.59% at a SPY beta of 0.51. Eighteen single-knob perturbations of the configuration, each reducing out-of-sample Sharpe, indicate that the contribution is structural rather than parameter-specific.

The full implementation, configuration files, fold checkpoints, and figure-generation scripts are publicly available at <https://github.com/Tyler-Pellek/regime-aware-allocation>.

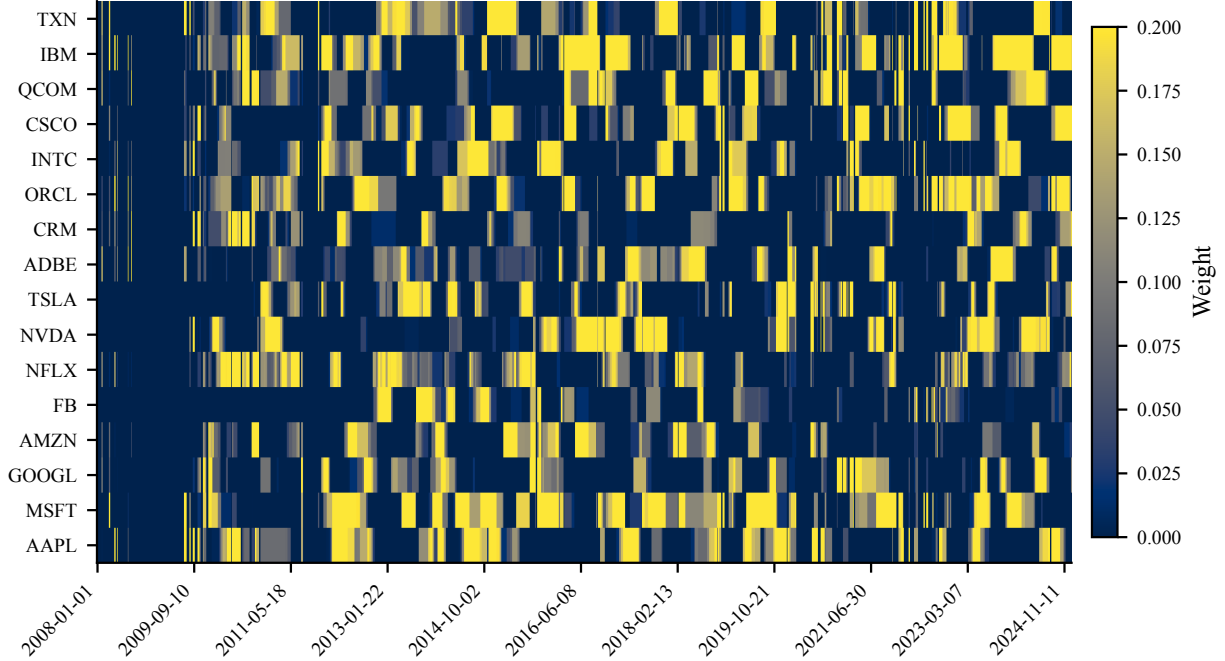


Figure 5: Out-of-sample portfolio weights through time (vertical axis: ticker, horizontal axis: rebalance date). Per-asset cap $w_{\max} = 0.20$; average active positions per rebalance is 6.72; annualized L1 turnover is 10.58.

A Ablation study

Each row of Tables 2–4 varies a single architectural, training, or optimization choice relative to the production configuration, evaluated on the same 4,417-day out-of-sample window. All runs are trained from scratch under the corresponding setting; no parameters are transferred between configurations.

Table 2: Mean-signal substitutions (input to the optimizer’s expected-return vector).

Mean signal	CAGR	Sharpe	MaxDD	Calmar
Rolling 60-day sample mean (production)	0.2356	1.21	−0.251	0.94
12-1 Jegadeesh–Titman momentum	0.1375	0.78	−0.455	0.30
0.5· mean + 0.5· 12-1 momentum	0.1365	0.80	−0.368	0.37
0.8· mean + 0.2· 12-1 momentum	0.1729	0.94	−0.393	0.44
0.9· mean + 0.1· 12-1 momentum	0.1886	1.00	−0.400	0.47
0.95· mean + 0.05· 12-1 momentum	0.1912	1.01	−0.408	0.47
63-day Sharpe momentum, cross-sectional z -score	0.1382	0.84	−0.393	0.35

Across the eighteen modifications, no configuration produces an out-of-sample Sharpe exceeding the production value of 1.21. The configurations with the smallest gap are the 5-seed ensemble (Sharpe 1.07) and the stronger-regularization variant (Sharpe 1.06), both of which exhibit maximum drawdowns larger than the production configuration. The configurations that substantially perturb the optimizer formulation (scenario-based CVaR, long-short, momentum signals at weight ≥ 0.20) reduce Sharpe more aggressively.

Table 3: Optimizer-formulation substitutions.

Optimizer variation	CAGR	Sharpe	MaxDD	Calmar
Mean-variance with vol cap (production)	0.2356	1.21	-0.251	0.94
Long-short, $w \in [-0.05, 0.20]$, gross ≤ 1.2	0.1984	1.03	-0.407	0.49
Long-short, $w \in [-0.10, 0.20]$, gross ≤ 1.5	0.1980	1.00	-0.416	0.48
Scenario-based CVaR, bootstrap with $M = 500$	0.1265	0.67	-0.479	0.26
Cross-validated risk aversion γ per fold	0.1837	1.00	-0.421	0.44
Anchor regularizer toward sample covariance	0.2261	1.03	-0.375	0.60

Table 4: Architectural and training-procedure modifications.

Modification	CAGR	Sharpe	MaxDD	Calmar
Production (warm-start + factor head + features)	0.2356	1.21	-0.251	0.94
Cold initialization (no warm-start)	0.0104	0.27	-0.099	0.10
Standard Cholesky head (not factor)	0.0840	1.02	-0.146	0.58
Universe-wide pretraining on 200 tickers	0.2139	1.06	-0.381	0.56
5-seed ensemble, Σ -averaged at inference	0.2029	1.07	-0.342	0.59
Realized-covariance target ($h=20$) + ensemble	0.1285	0.92	-0.299	0.43
Stronger regularization (dropout 0.25, wd 1.5×10^{-3})	0.1901	1.06	-0.341	0.56
Per-asset temporal attention encoder (weekly train)	0.1746	0.91	-0.367	0.48
Per-asset temporal attention encoder (daily train)	0.1914	0.91	-0.398	0.48

B Hyperparameter settings

Table 5 reports the production hyperparameter configuration. Values were selected on the first fold’s training and in-fold validation windows and held fixed for the remaining seventeen folds; no hyperparameter is tuned on out-of-sample data. The transformer width ($d_{\text{model}} = 192$), depth (5 layers), and factor rank ($k = 4$) trade parameter count against the modest sample size of 60-day windows: the model has approximately 1.8 million parameters and trains in under five minutes per fold on a single A100. The Cholesky diagonal floor $\sigma_{\min} = 10^{-2}$ is the smallest value at which the covariance head remained numerically stable under FP32 throughout backtesting; values an order of magnitude smaller produced occasional non-positive-definite outputs in the validation set. The risk-aversion coefficient $\gamma = 25$, the weekly volatility cap $\bar{\sigma} = 0.040$, the per-asset cap $w_{\max} = 0.20$, and the inference shrinkage $\alpha = 0.70$ are the four levers with the largest impact on realized risk; ablations of each are reported in Table 3. The walk-forward schedule uses an initial training cutoff of 2007-12-31 so that the 2008 global financial crisis is fully out-of-sample for fold one, fold test windows of 252 trading days, weekly rebalances (5 trading days), and a 5-day training snapshot stride.

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Table 5: Production hyperparameter settings.

Component	Hyperparameter	Value
Transformer	Embedding dimension d_{model}	192
	Attention heads	8
	Layers	5
	Feed-forward expansion	4
	Dropout	0.15
	Cholesky diagonal floor σ_{min}	10^{-2}
Factor head	Latent factors k	4
	Diagonal-shift initialization c	4.0
Training	Lookback window W	60
	Forward horizon h	5
	Batch size	32
	Max epochs	30
	Learning rate (AdamW)	3×10^{-4}
	Weight decay	5×10^{-4}
	Gradient clipping norm	1.0
	Warm-start LR scale η_{warm}	0.5
Optimizer	Risk aversion γ	25.0
	Turnover penalty λ_{TO}	5×10^{-3}
	Per-asset cap w_{max}	0.20
	Weekly vol cap $\bar{\sigma}$	0.040
	Inference shrinkage α	0.70
	Solver	Clarabel
Regime model	Hidden states K	3
	Covariance type	Full
	Re-fit cadence	Per fold
Walk-forward	Initial train cutoff	2007-12-31
	Fold test length (days)	252
	Rebalance frequency (days)	5
	Train snapshot stride (days)	5

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